A Course Mini Project report on

Student Result Processing System

Submitted inpartial fulfilment ofthe requirements.

for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING

by

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SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY
(AUTONOMOUS) ANANTHAPURAMU

(Affiliated to JNTUA & Approved by AICTE) (Accredited by NAAC With 'A'Grade & Accredited by NBA (EEE, ECE & CSE))

2025-2026

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Certificate

This is to certify that the Course Mini project report entitled Student Result Processing System is the bona fide work carried out by **P.Pallavi** bearing Roll Number **244G1A05S0** in partial fulfilment of the requirements for the CAA during the academic year 2025-2026.

Signature of Guide

Mr.L.Suman M.Tech

Assistant Professor

Head of the Department

Dr.P.Veera Prakash M.Tech., Ph.D

Assistant Professor

Date:

Place: Ananthapuramu

DECLARATION

We hereby declare that the course mini project report entitled "**Student Result Processing System**" submitted for the CAA is our original work.

Date:

Place: :Ananthapuramu

Course Mini Project Associate

P.Pallavi

244G1A05S0

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without the mention of people who made it possible, whose constant guidance and encouragement crowned our efforts with success. It is a pleasant aspect that we have now the opportunity to express my gratitude for all of them.

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Course Mini Project Associate

P.Pallavi

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ABSTRACT

The Student Result Processing System is a Java-based software application developed to simplify and automate the process of managing student academic results. In traditional systems, result processing is often done manually, which is time-consuming and prone to human errors. This project aims to overcome these limitations by providing an efficient, accurate, and reliable computerized solution using Core Java.

The system allows the user to enter student details such as roll number, name, class, and marks obtained in multiple subjects. By applying Object-Oriented Programming principles like classes, objects, encapsulation, and methods, the application processes student data in an organized and structured manner. Java control statements such as if-else, loops, and switch-case are used to calculate total marks, percentage, grades, and to determine pass or fail status based on predefined academic rules.

The application ensures quick result generation with minimal manual intervention. It reduces calculation errors and maintains consistency in grading. The system also supports easy storage, retrieval, and display of student records, enabling teachers and administrators to analyze student performance efficiently. This project highlights the practical use of Java concepts such as arrays, Scanner class for input handling, and modular programming techniques.

Overall, the Student Result Processing System improves efficiency, transparency, and accuracy in academic result management. It serves as an effective learning project for understanding Java programming and Object-Oriented concepts and can be further enhanced by integrating database connectivity and graphical user interfaces in future developments.

Keywords: Student Result Processing System, Java, Core Java, Object-Oriented Programming (OOP), Classes and Objects, Marks Calculation, Grade Evaluation, Pass/Fail Analysis, Data Management, Control Statements, Arrays, Scanner Class, Academic Performance, Automation, Educational Software

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Chapter 1
Introduction

1.1 Overview

The Student Result Processing System collects student details such as roll number, name, and marks obtained in various subjects. It processes this data to calculate total marks, percentage, grades, and pass or fail status based on predefined rules. The system provides a simple and user-friendly interface that allows teachers and administrators to generate results quickly and reliably. It helps maintain consistency, transparency, and accuracy in academic result evaluation. The Student Result Processing System provides a structured approach to handling student academic data by automating result computation and record management. The system ensures uniform evaluation criteria for all students and maintains data integrity. It allows authorized users to input, update, and view student results efficiently. By using Java programming, the system follows a modular design, making it easy to understand, maintain, and enhance. This application acts as a reliable tool for educational institutions to manage large volumes of student data with accuracy and speed.

1.2 Objective

- To automate the student result calculation process
- To reduce manual effort and human errors in result preparation
- To efficiently store and manage student academic records
- To generate accurate results including grades and pass/fail status
- To demonstrate the practical application of Java and OOP concepts.
- To ensure consistency and fairness in result evaluation.
- To minimize redundancy and duplication of student records.

1.3 Scope

The scope of the Student Result Processing System includes managing student data and academic results within educational institutions such as schools and colleges. The system is designed for small to medium-scale usage and can be extended in the future by integrating database support, graphical user interfaces, and advanced reporting features. It serves as an effective learning tool for understanding Java programming and can be enhanced to meet real-world academic requirements. The scope of this project can be expanded to include database integration, online result publishing, role-based access control, and performance analytics. It can also be extended to support web-based and mobile applications, making result management more accessible and user-friendly in the future.

Chapter 2

Literature Review

2.1 Manual Result Processing Systems

Traditional student result processing was done manually using paper records and registers. These methods were time-consuming, error-prone, and difficult to manage for large numbers of students. In earlier educational systems, student results were maintained using paper records, registers, and manual calculations. These methods required a lot of time and effort from teachers and administrators.

2.2 Computerized Result Management Systems

With the introduction of computer-based systems, result processing became faster and more accurate. These systems reduced manual errors and improved data storage, but some lacked flexibility and scalability. With the introduction of computers, result processing systems were developed using basic software tools and spreadsheets.

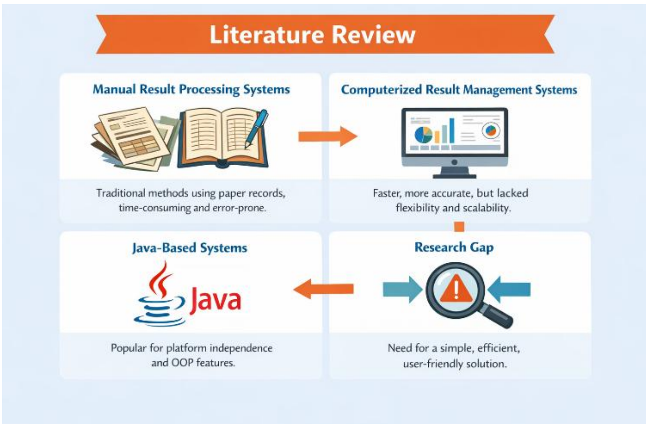

Fig 2.1.1. Literature Review

2.3 Java-Based Systems

Java-based result processing systems gained popularity due to Java's platform independence and object-oriented features. Such systems provide modular design, better data management, and easy system enhancement. Studies show that Java-based systems provide better performance, data security, and flexibility.

2.4 Research Gap

Existing systems highlight the need for a simple, efficient, and user-friendly Student Result Processing System. The proposed Java-based system aims to overcome limitations of earlier methods by improving accuracy, reliability, and performance.Although several result management systems exist, many are complex, costly, or difficult to use. There is a need for a simple, efficient, and user-friendly Student Result Processing System that ensures accuracy and reliability.

Chapter 3
System Design

3.1 System Architecture

The Student Result Processing System follows a simple modular architecture. It consists of input, processing, and output modules. The input module collects student details and subject marks. The processing module calculates total marks, percentage, grades, and pass/fail status using predefined rules. The output module displays the final result in a readable format. This structured design improves clarity, efficiency, and maintainability.

3.2 Input Design

The input design focuses on accurate and user-friendly data entry. The system accepts inputs such as student roll number, name, and marks for each subject. Input validation is performed to ensure that only valid marks are entered. Using the Scanner class in Java, the system ensures smooth and error-free data collection.

3.3 Processing Design

The processing design handles all calculations related to student results. It uses Java control statements like loops and conditional statements to compute total marks, average, percentage, and grades. Object-Oriented Programming concepts such as classes and methods are used to organize the processing logic, making the system modular and reusable.

3.4 Output Design

The output design presents the processed results clearly. The system displays student details along with total marks, percentage, grade, and pass/fail status. The output is designed in a simple and understandable format to ensure easy interpretation by users.

3.5 Security and Validation

The system includes basic validation to prevent incorrect data entry. Marks are checked to ensure they are within allowed limits. This helps maintain data integrity and reliability of results.

3.6 Data Flow Design

The data flow starts with user input, followed by result processing, and ends with result display. Student data flows smoothly between modules without redundancy. This design ensures accuracy and consistency throughout the system.

Chapter 4
Implementation & output

Source code:

```
import java.util.*;
import java.io.*;

// Interface
interface ResultOperations
 {
   void calculateResult();
   void display();
}

// Base Class
class Person
 {
   protected String name;
   protected int rollNo;

   public Person(String name, int rollNo)
{
      this.name = name;
      this.rollNo = rollNo;
   }
}

// Derived Class
class Student extends Person implements ResultOperations

 {
   private int[] marks;
   private int total;
   private double percentage;
   private String result;
   private static int studentCount = 0;
```

```
    public Student(String name, int rollNo)
  {
        super(name, rollNo);
        marks = new int[6];    // Fixed 6 subjects

 studentCount++;
    }

    public void inputMarks(Scanner sc)
  {
        total = 0;
        for (int i = 0; i < 6; i++)
  {
            while (true)
 {
                Try
  {
                    System.out.print("Subject " + (i + 1) + " (0-100): ");
                    int m = sc.nextInt();
                    if (m < 0 || m > 100)
                        throw new IllegalArgumentException();
                    marks[i] = m;
                    total += m;
                    break;
                }

catch (Exception e)
{
                    System.out.println("Invalid input! Enter marks between 0-100.");
                    sc.nextLine();
                }
            }
        }
        calculateResult();
    }

    public void calculateResult()
```

```
{
        percentage = total / 6.0;
        result = "PASS";
        for (int m : marks)
 {
            if (m < 35) {
                result = "FAIL";


 break;
            }
        }
    }

    public void display()
{
        System.out.println("--------------------------------");
        System.out.println("Name : " + name);
        System.out.println("Roll No : " + rollNo);
        System.out.println("Total : " + total);
        System.out.println("Percentage : " + percentage);
        System.out.println("Result : " + result);
        System.out.println("--------------------------------");
    }

    public String getResult()
 {
        return result;
    }

    public int getRollNo()
{
        return rollNo;
    }

    public static int getStudentCount()
{
        return studentCount;
```

```
  }
   public void saveToFile()
{
       Try
 {
           FileWriter fw = new FileWriter("results.txt", true);
           fw.write(name + " | " + rollNo + " | " + percentage + " | " + result + "\n");
           fw.close();
       }
catch (IOException e)
 {
           System.out.println("Error saving to file.");
       }
   }
}

// Main Class
public class StudentProcess
{
   public static void main(String[] args)
 {
       Scanner sc = new Scanner(System.in);
       ArrayList<Student> students = new ArrayList<>();

       int
choice;
       do {
           System.out.println("\n===== RESULT PROCESSING MENU =====");
           System.out.println("1. Add Student");
           System.out.println("2. Display All Students");
           System.out.println("3. Search by Roll Number");
           System.out.println("4. Show Student Count");
           System.out.println("5. Display PASS Students");
           System.out.println("6. Display FAIL Students");
           System.out.println("7. Exit");
           System.out.print("Enter your choice: ");
```

```
 choice = sc.nextInt();

          switch (choice)
 {

              case 1:
                 sc.nextLine();
                 System.out.print("Enter Name: ");
                 String name = sc.nextLine();

                 System.out.print("Enter Roll Number: ");
                 int roll = sc.nextInt();

                 Student s = new Student(name, roll);
                 s.inputMarks(sc);
                 s.saveToFile();
                 students.add(s);
                 break;

              case 2:
                 if (students.isEmpty())
{
                    System.out.println("No students available.");
                 }
else
{
                    for (Student st : students)
                       st.display();
                 }
                 break;

              case 3:
                 System.out.print("Enter Roll Number to search: ");
                 int search = sc.nextInt();
                 boolean found = false;

                 for (Student st : students)
```

```
 {
            if (st.getRollNo() == search)
{
                st.display();
                found = true;
                break;
            }
        }

        if (!found)
            System.out.println("Student not found!");
        break;

    case 4:
        System.out.println("Total Students: " + Student.getStudentCount());
        break;

    case 5:
        System.out.println("\n--- PASS STUDENTS ---");
        boolean passFound = false;
        for (Student st : students) {
            if (st.getResult().equals("PASS"))
{
                st.display();
                passFound = true;
            }
 }
        if (!passFound)
            System.out.println("No PASS students found.");
        break;

    case 6:
        System.out.println("\n--- FAIL STUDENTS ---");
        boolean failFound = false;
        for (Student st : students) {
            if (st.getResult().equals("FAIL"))
 {
```

```
st.display();
                    failFound = true;
                }
            }
            if (!failFound)
                System.out.println("No FAIL students found.");
            break;

        case 7:
            System.out.println("Exiting system...");
            break;

        default:
            System.out.println("Invalid choice!");
        }

    } while (choice != 7);

    sc.close();
  }
}
```

OUTPUT:

```
===== RESULT PROCESSING MENU =====
1. Add Student
2. Display All Students
3. Search by Roll Number
4. Show Student Count
5. Display PASS Students
6. Display FAIL Students
7. Exit
Enter your choice: 1
Enter Name: P.Pallavi
Enter Roll Number: 123
Subject 1 (0-100): 56
Subject 2 (0-100): 7
Subject 3 (0-100): 76
Subject 4 (0-100): 78
Subject 5 (0-100): 87
Subject 6 (0-100): 86

===== RESULT PROCESSING MENU =====
1. Add Student
2. Display All Students
3. Search by Roll Number
4. Show Student Count
5. Display PASS Students
6. Display FAIL Students
7. Exit
Enter your choice: 1
```

```
Enter Name: C.Padmini
Enter Roll Number: 124
Subject 1 (0-100): 67
Subject 2 (0-100): 90
Subject 3 (0-100): 56
Subject 4 (0-100): 87
Subject 5 (0-100): 89
Subject 6 (0-100): 56

===== RESULT PROCESSING MENU =====
1. Add Student
2. Display All Students
3. Search by Roll Number
4. Show Student Count
5. Display PASS Students
6. Display FAIL Students
7. Exit
Enter your choice: 1
Enter Name: P.Rupa
Enter Roll Number: 122
Subject 1 (0-100): 87
Subject 2 (0-100): 87
Subject 3 (0-100): 86
Subject 4 (0-100): 67
Subject 5 (0-100): 56
Subject 6 (0-100): 89
```

```
===== RESULT PROCESSING MENU =====
1. Add Student
2. Display All Students
3. Search by Roll Number
4. Show Student Count
5. Display PASS Students
6. Display FAIL Students
7. Exit
Enter your choice: 1
Enter Name: L.Nnandini
Enter Roll Number: 126
Subject 1 (0-100): 78
Subject 2 (0-100): 86
Subject 3 (0-100): 44
Subject 4 (0-100): 76
Subject 5 (0-100): 68
Subject 6 (0-100): 89

===== RESULT PROCESSING MENU =====
1. Add Student
2. Display All Students
3. Search by Roll Number
4. Show Student Count
5. Display PASS Students
6. Display FAIL Students
7. Exit
Enter your choice: 1
```

```
Enter Name: D.Meghana
Enter Roll Number: 129
Subject 1 (0-100): 66
Subject 2 (0-100): 90
Subject 3 (0-100): 78
Subject 4 (0-100): 80
Subject 5 (0-100): 68
Subject 6 (0-100): 70

===== RESULT PROCESSING MENU =====
1. Add Student
2. Display All Students
3. Search by Roll Number
4. Show Student Count
5. Display PASS Students
6. Display FAIL Students
7. Exit
Enter your choice: 2
--------------------------------
Name : P.Pallavi
Roll No : 123
Total : 390
Percentage : 65.0
Result : FAIL
--------------------------------
--------------------------------
Name : C.Padmini
Roll No : 124
Total : 445
Percentage : 74.16666666666667
Result : PASS
--------------------------------
```

```
Name : P.Rupa
Roll No : 122
Total : 472
Percentage : 78.66666666666667
Result : PASS
--------------------------------
--------------------------------
Name : L.Nnandini
Roll No : 126
Total : 441
Percentage : 73.5
Result : PASS
--------------------------------
--------------------------------
Name : D.Meghana
Roll No : 129
Total : 452
Percentage : 75.33333333333333
Result : PASS
--------------------------------

===== RESULT PROCESSING MENU =====
1. Add Student
2. Display All Students
3. Search by Roll Number
4. Show Student Count
5. Display PASS Students
6. Display FAIL Students
7. Exit
Enter your choice: 3
Enter Roll Number to search: 125
Student not found!
```

```
===== RESULT PROCESSING MENU =====
1. Add Student
2. Display All Students
3. Search by Roll Number
4. Show Student Count
5. Display PASS Students
6. Display FAIL Students
7. Exit
Enter your choice: 3
Enter Roll Number to search: 122
--------------------------------
Name : P.Rupa
Roll No : 122
Total : 472
Percentage : 78.66666666666667
Result : PASS
--------------------------------

===== RESULT PROCESSING MENU =====
1. Add Student
2. Display All Students
3. Search by Roll Number
4. Show Student Count
5. Display PASS Students
6. Display FAIL Students
7. Exit
Enter your choice: 4
Total Students: 5
```

```
===== RESULT PROCESSING MENU =====
1. Add Student
2. Display All Students
3. Search by Roll Number
4. Show Student Count
5. Display PASS Students
6. Display FAIL Students
7. Exit
Enter your choice: 5

--- PASS STUDENTS ---
--------------------------------
Name : C.Padmini
Roll No : 124
Total : 445
Percentage : 74.16666666666667
Result : PASS
--------------------------------
--------------------------------
Name : P.Rupa
Roll No : 122
Total : 472
Percentage : 78.66666666666667
Result : PASS
--------------------------------
--------------------------------
Name : L.Nnandini
Roll No : 126
Total : 441
Percentage : 73.5
Result : PASS
--------------------------------
```

```
Name : D.Meghana
Roll No : 129
Total : 452
Percentage : 75.33333333333333
Result : PASS
--------------------------------

===== RESULT PROCESSING MENU =====
1. Add Student
2. Display All Students
3. Search by Roll Number
4. Show Student Count
5. Display PASS Students
6. Display FAIL Students
7. Exit
Enter your choice: 6

--- FAIL STUDENTS ---
--------------------------------
Name : P.Pallavi
Roll No : 123
Total : 390
Percentage : 65.0
Result : FAIL
--------------------------------
```

```
===== RESULT PROCESSING MENU =====
1. Add Student
2. Display All Students
3. Search by Roll Number
4. Show Student Count
5. Display PASS Students
6. Display FAIL Students
7. Exit
Enter your choice: 7
Exiting system...
```

Conclusion

The Student Result Processing System is an effective application developed using Java to automate the process of evaluating student performance. It simplifies the calculation of total marks, average, percentage, and grade by using programming logic instead of manual computation. This reduces human errors and saves time for teachers and institutions.

The system demonstrates important object-oriented programming concepts such as classes, objects, constructors, arrays, loops, and conditional statements. It provides a clear understanding of how real-world problems can be solved using Java programming.

Moreover, this system can be expanded further by adding features such as:

- Storing student records in files or databases
- Processing multiple students at a time
- Generating report cards
- Adding subject-wise pass/fail status
- Creating a graphical user interface (GUI)

In conclusion, the Student Result Processing System is a practical and foundational project that enhances programming skills and shows how technology can improve efficiency in educational institutions.

References

- Java: The Complete Reference – Herbert Schildt
- Used for understanding core Java concepts, classes, objects, arrays, and control statements.
- Head First Java – Kathy Sierra and Bert Bates
- Helpful for learning object-oriented programming concepts in an easy way.
- Oracle Corporation – Java Documentation
- Official reference for Java syntax, Scanner class, and program structure.
- College lecture notes and classroom materials on Java programming.
- Online tutorials and practice examples related to student result processing programs.

Evaluation Parameters

Criteria	Marks	Marks Obtained
Problem Definition	2	
Design & Methodology	2	
Implementation & coding	2	
Presentation	2	
Report	2	

Signature of the Faculty **Signature of the HOD**